

Impact of Tropical Cyclone ‘Asna’ on Solar Energy Generation in Gujarat: A Meteorological Perspective

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Abstract

India’s growing renewable energy sector depends heavily on accurate weather forecasting. This study examines how Cyclone Asna (August 25–30, 2024) affected solar power generation in Gujarat during an anomalous monsoon season. Using ERA-5 reanalysis data, we analyzed Global Horizontal Irradiance (GHI), precipitation, and cloud cover from cyclogenesis to dissipation. Despite Monsoon conditions being unfavorable for Cyclogenesis Asna intensified over land. During these time Gujarat Experienced wet to severe wet conditions with 80% decrease in solar generation due to diminished GHI exposing renewable energy’s vulnerability to extreme weather conditions. This case study emphasizes the importance of integrating high-resolution, energy-focused weather models into operational forecasting frameworks to support grid reliability and market resilience in the face of increasing climate variability.

Introduction

Renewable energy is one of the major applications of operational meteorology, the slight fluctuations in the localized weather even can cause a major loss in the solar generation. The study shows the significance of meteorology in the renewable energy sector. Here we studied how the weather impacted the solar energy generation of northwest India during the last year (2024) and topical cyclone ASNA which drastically influenced the solar and wind generation of northwest India during peak southwest monsoon.

Data and Methodology

- ERA-5 - 1991 to 2024 with 0.25° x 0.25° resolution for rainfall, SST, wind speed at 100m and total cloud cover.
- IBTrACS: Tropical cyclone best track data for ASNA.
- IMD : Yearly Gridded Rainfall (0.25° x 0.25°).
- During the monsoon season, 13 low-pressure systems (one cyclonic storm, 3 Deep Depressions, 2 Depressions, 2 well-marked low-pressure areas, 4 low-pressure areas and one low-pressure area over land) were formed.
- 2024 IMD annual report shows the year as an anomalous year for the rainfall and temperature.
- Saurashtra and Kutch regions, witnessed a positive rainfall anomaly exceeding 400 mm.
- JJAS (Jun,Jul,Aug,Step) rainfall anomaly shows positive anomaly over Saurashtra and Kutch regions with more than 300mm.
- The northeast regions with a negative anomaly of more than 400 mm.

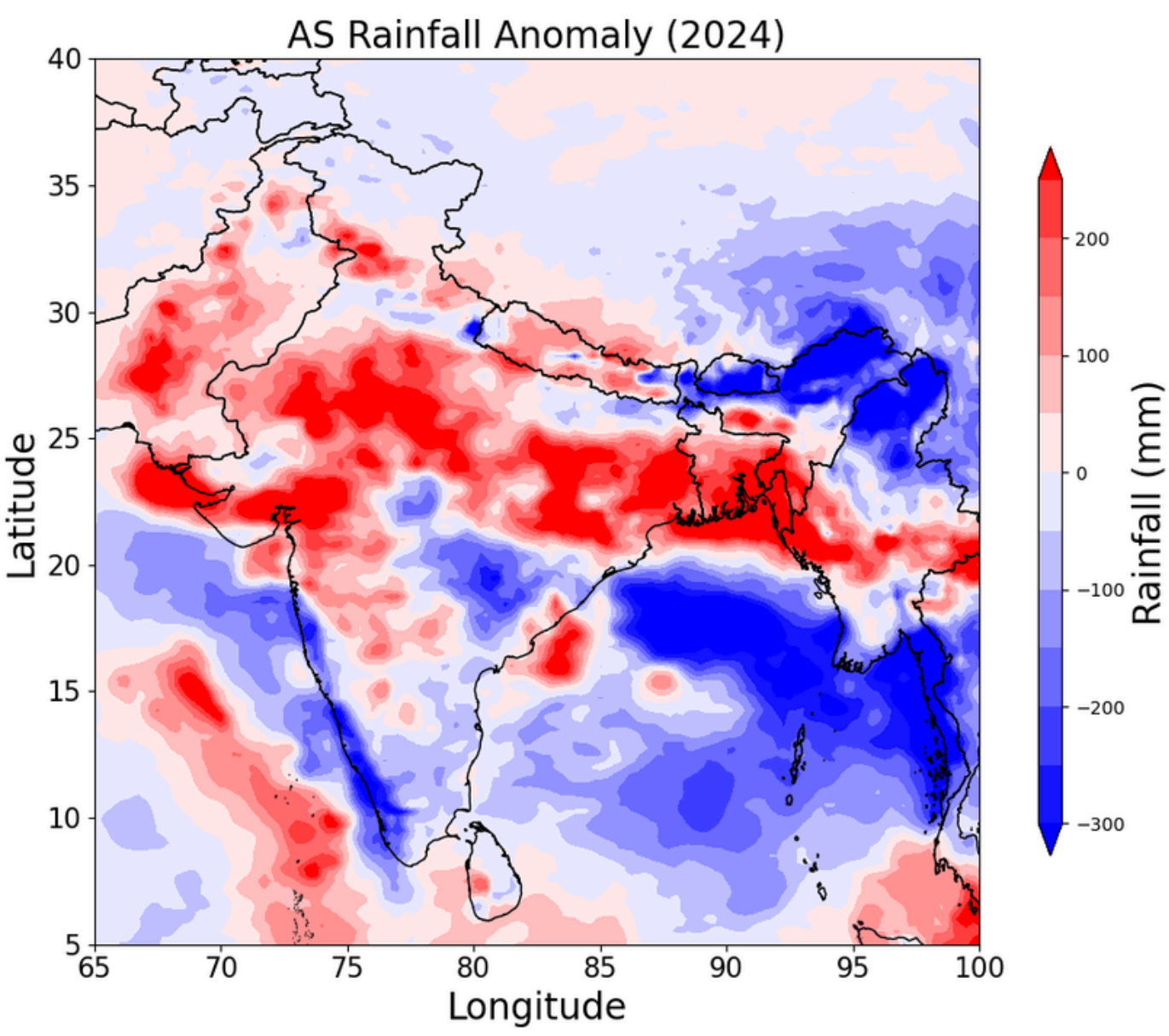
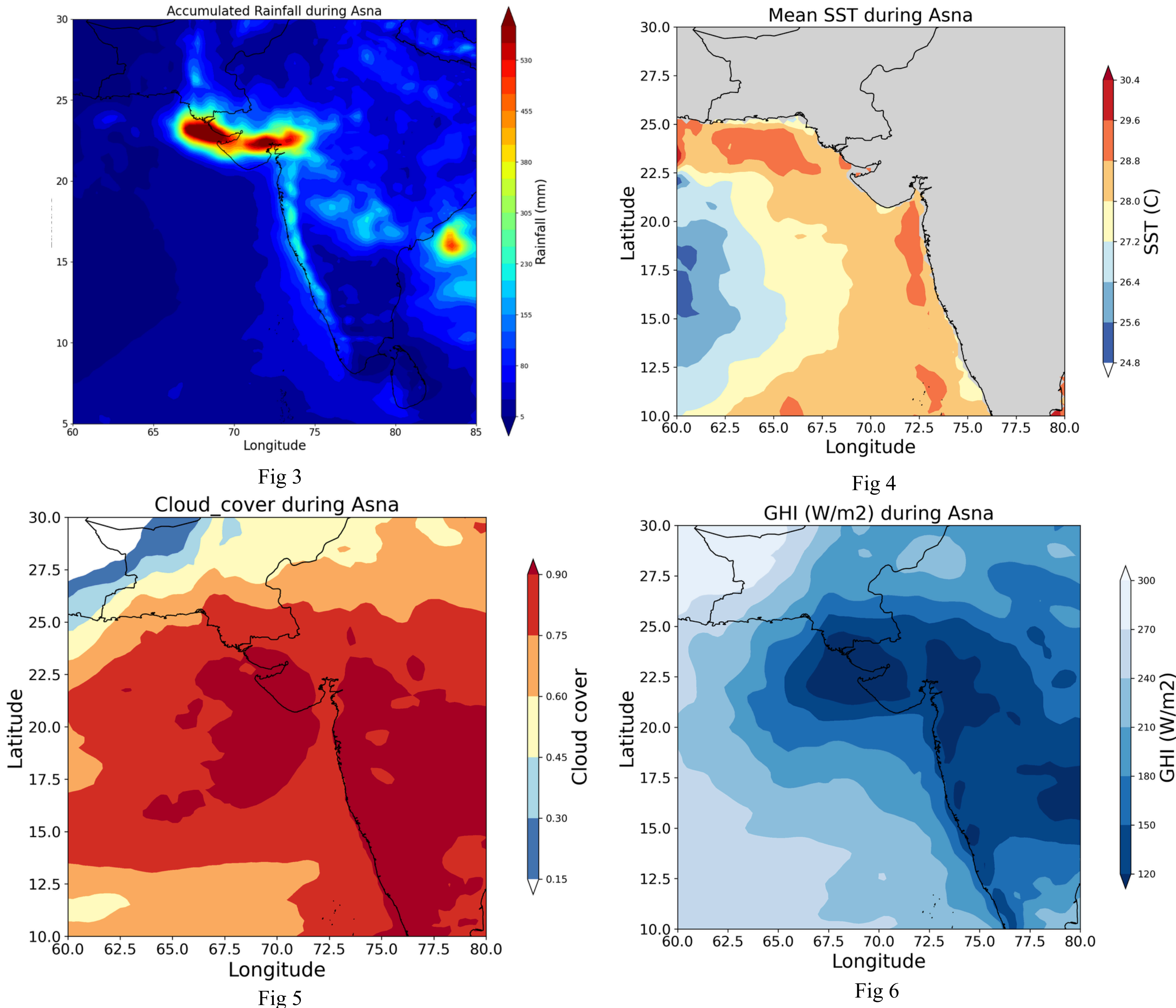


Fig 1



The Fig 3 shows the nine days accumulated rainfall over the the west coast of India during the Asna cyclone. The Saurashtra and Kutch regions exhibits an accumulated rainfall of above 600mm during this period. (Fig4) An excess in the SST is also observed in the Gujarat coast during the same period. (Fig5) The GHI over the Gujarat region also shows decrease in GHI which is unfavored for solar energy generation.

Why ASNA ?

TROPICAL cyclone (TC) is a multi-hazardous natural weather phenomenon associated with strong wind, heavy rainfall and storm surge. Asna developed over the AS during the peak monsoon is rare, where the meteorological conditions are not favorable for a cyclogenesis. TC developed in the month of July formed over the northeast Arabian Sea and made landfall over the Gujarat and Saurashtra coasts.

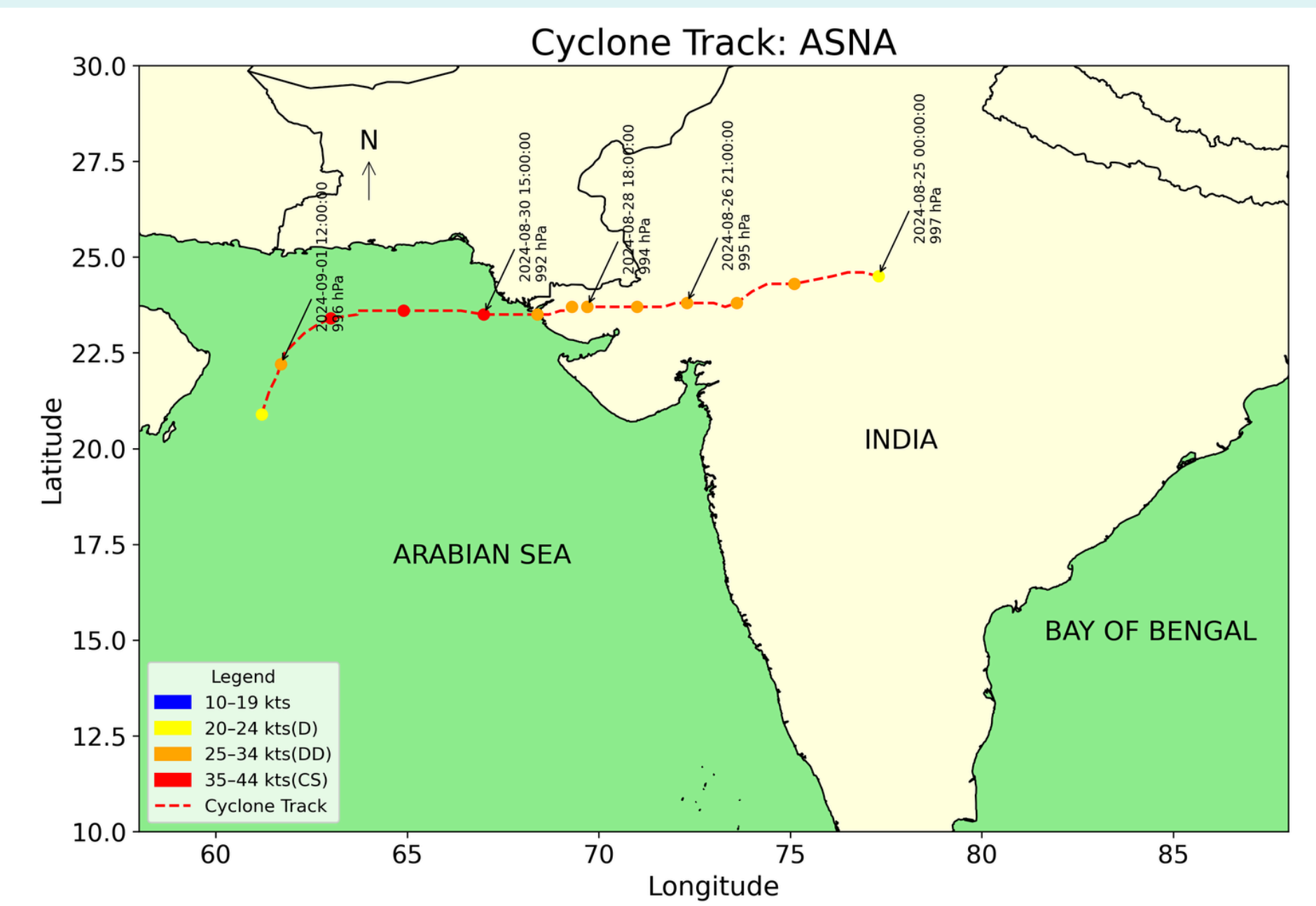


Fig 2

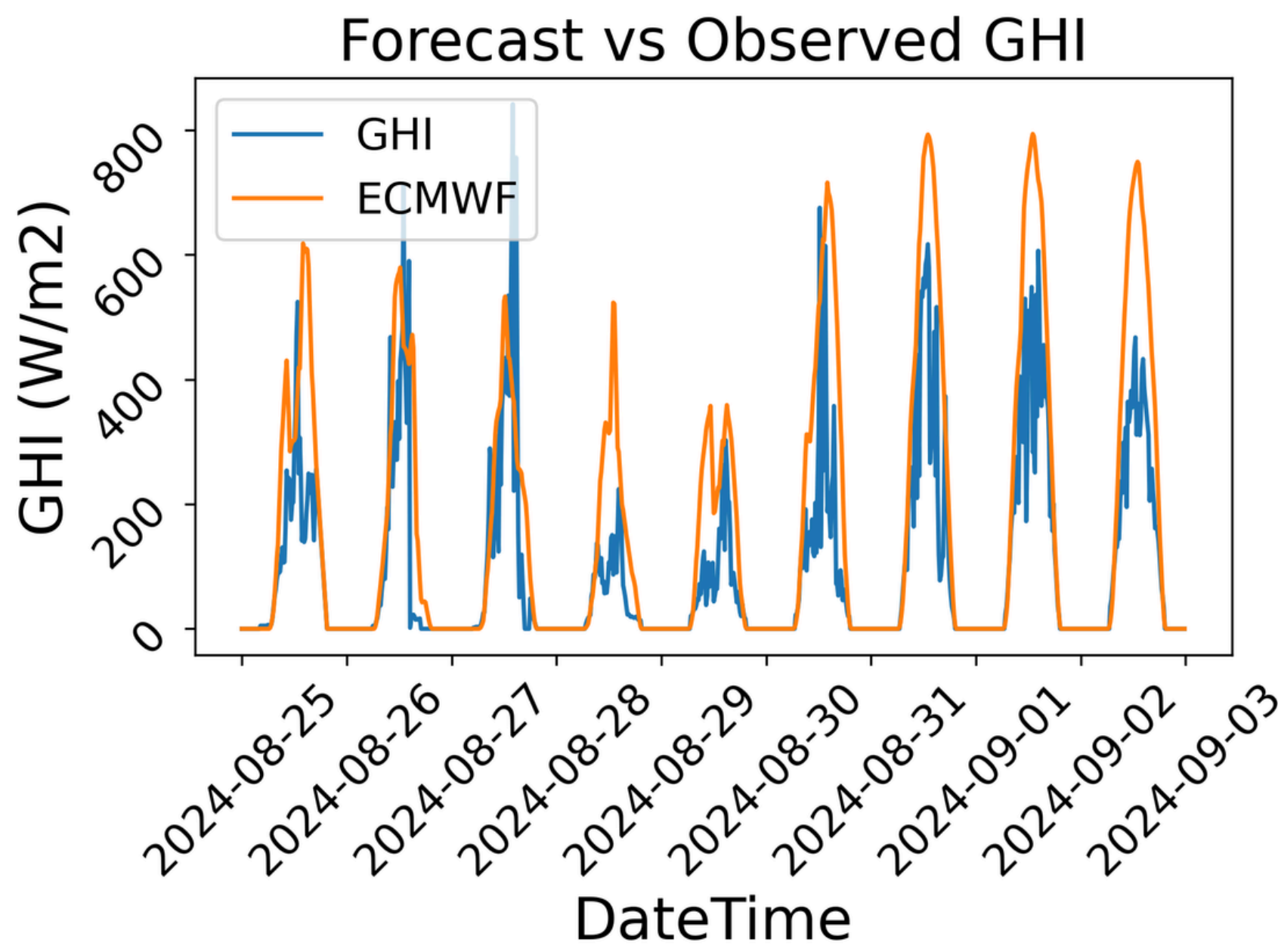


Fig 7

- The figure shows the time series comparison between ECMWF forecast and the observed GHI during the Asna cyclone intensification period.
- The models failed to exactly predict the fall in GHI by the clouding due to the cyclone outskirt.
- The more fall in GHI is observed during 27th to 30th Aug where the deep depression was there over the Gujarat region.
- And it further intensified as cyclone during 30th Aug and moved towards the Arabian Sea which caused the rise in GHI after 30th Aug.
- The figure shows a forecasting Mean Absolute Error (MAE) of 79.87%.
- Root Mean Square error of 147.47 W/m² on an average from observed GHI.

Results and Discussions

- The uneven and sudden changes in the weather is directly impacting the renewable energy sector and there after by the stability of the Indian electricity Grid.
- The rare case of Tropical Cyclone Asna also impacted the renewable energy sector in a drastic level.
- The Global and regional NWP models were unable to predict the variations in GHI and cloud cover during the period.
- Here it shows the urgent need of NWP models which specifically focused on renewable energy Sector (solar and wind) for the better operational procedures and for the Growth of renewable sector in the country. IoT sensors can be used for collecting more accurate data and for better predictions.

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Long-Term Trends and Monsoon Anomalies in Global Horizontal Irradiance over Key Renewable Energy Zones in India



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Abstract

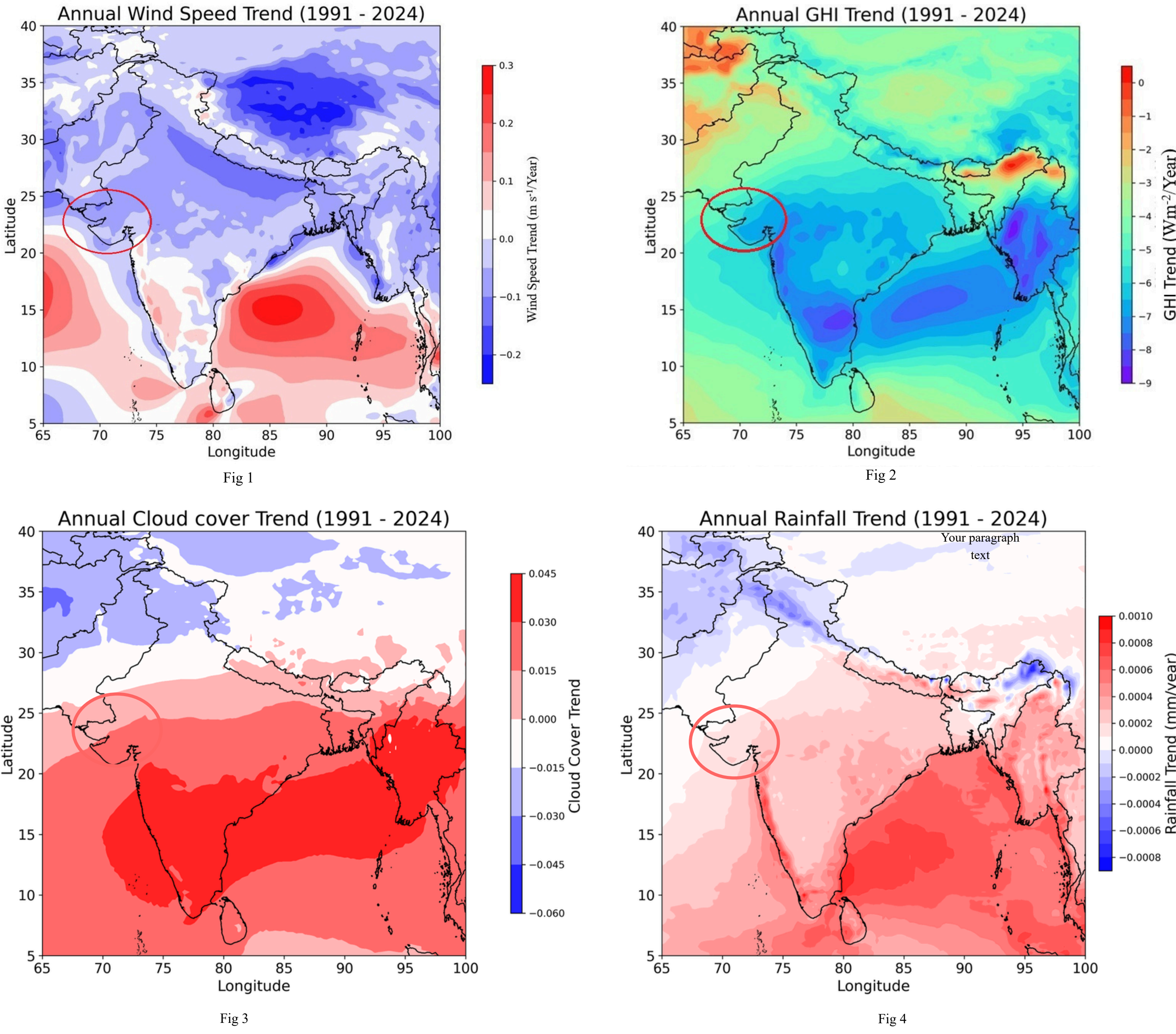
Renewable energy sector of India is predominantly driven by solar and wind resources, with minimal carbon footprint. Adani Green Energy Limited currently holds a renewable energy portfolio of approximately 16 GW, with operational forecasts which are heavily reliant on solar and wind resource assessments. This study investigates the long-term climatological trends of Global Horizontal Irradiance (GHI)—a critical parameter representing surface solar radiation reaching photovoltaic panels—using hourly ERA5 reanalysis data at a spatial resolution of $0.25^\circ \times 0.25^\circ$. We conducted a 20-year trend analysis (1991–2024) of GHI across four strategic renewable energy hubs: Khavda (Gujarat), Jaisalmer (Rajasthan), Pavagada (Karnataka), and Kamuthi (Tamil Nadu). The analysis reveals a consistent declining trend in GHI across all selected regions. This reduction is attributed to increased rainfall and cloud cover, particularly during the monsoon season, which directly attenuates solar radiation. From a climate change perspective, rising temperature enhances water holding capacity of air, leading to strong convection and subsequent heavy precipitation, thereby diminishing solar irradiance. Further, we examined GHI anomalies and standard deviations for the monsoon seasons of 2024 and 2025. The results indicate a pronounced negative anomaly in 2025, especially over Gujarat and Rajasthan, suggesting a significantly wetter monsoon compared to 2024. Decadal trend analysis also supports the observed decline in solar radiation, reinforcing the need to integrate climatological variability into renewable energy forecasting models. This study underscores the importance of incorporating long-term and seasonal GHI variability into energy yield assessments and planning for solar infrastructure, especially under evolving climate conditions.

Introduction

India's renewable energy sector is largely powered by solar and wind, offering a minimal carbon footprint. Adani Green Energy Limited, with a 16 GW portfolio, relies heavily on accurate solar resource assessments. This study analyzes 30 years (1991–2024) of Global Horizontal Irradiance (GHI) trends using ERA5 reanalysis data across four key hubs—Khavda, Jaisalmer, Pavagada, and Kamuthi. Findings reveal evolving seasonal patterns driven by climate-induced shifts in rainfall and cloud cover, impacting solar performance during monsoon months. Short-term variability in GHI anomalies during 2024–2025 underscores the need for adaptive forecasting models. Integrating long-term and seasonal GHI variability into planning is critical for optimizing solar infrastructure under changing climate conditions.

Data and Methodology

This study investigates the long-term climatological trends of Global Horizontal Irradiance (GHI)—a critical parameter representing surface solar radiation reaching photovoltaic panels—using hourly ERA5 reanalysis data at a spatial resolution of $0.25^\circ \times 0.25^\circ$. We conducted a 20-year trend analysis (1991–2024) of GHI across four strategic renewable energy hubs: Khavda (Gujarat), Jaisalmer (Rajasthan), Pavagada (Karnataka), and Kamuthi (Tamil Nadu).



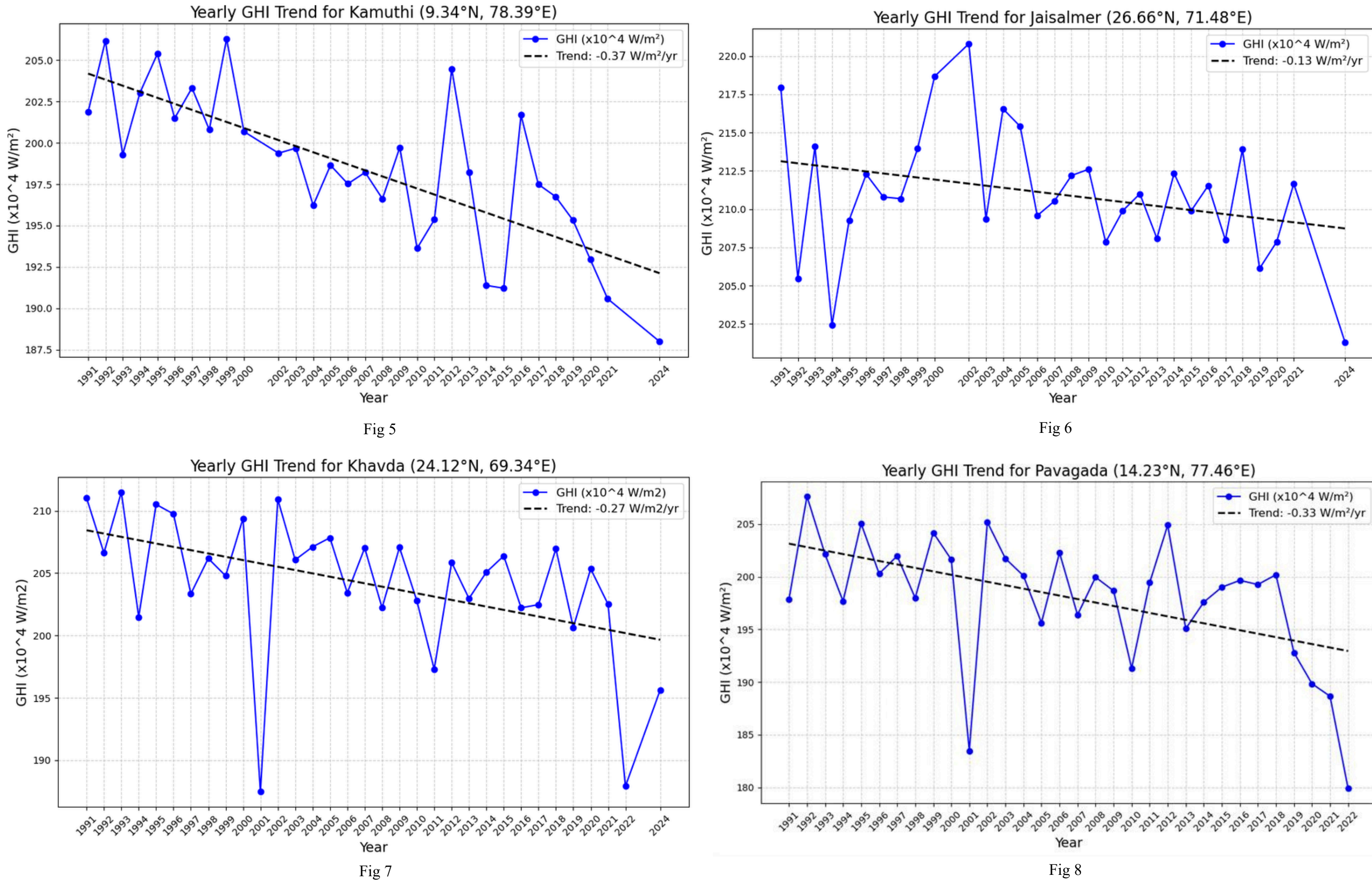
Annual trend for GHI during 1991-2024 shows a declining trend for solar energy all over India except Northeast region. This can be attributed to the extreme weather events which are occurring due to the global warming and thereby more rainfall and clouds over Indian region. Due to the global warming temperature of the atmosphere is increasing and then water holding capacity of the air is increasing thereby more cloud formation and more weather extremes.

Annual cloud cover trend also showing the similar signal to the GHI with increasing cloud cover over Indian region especially over central and south India. Over Bay of Bengal also the trend of cloud cover is increasing.

Annual rainfall trend also showing the similar signal to the GHI over long term. Rainfall is increasing over most parts of India except the Jammu Kashmir region and North eastern region. Over Bay of Bengal also the trend of rainfall is increasing because of the formation of more low pressure systems.

The wind speed (100 m) trend also found to be decreasing over different locations over India including Gujarat region. Over the Bay of Bengal due to the formation of more cyclonic systems, the wind speed trend is increasing.

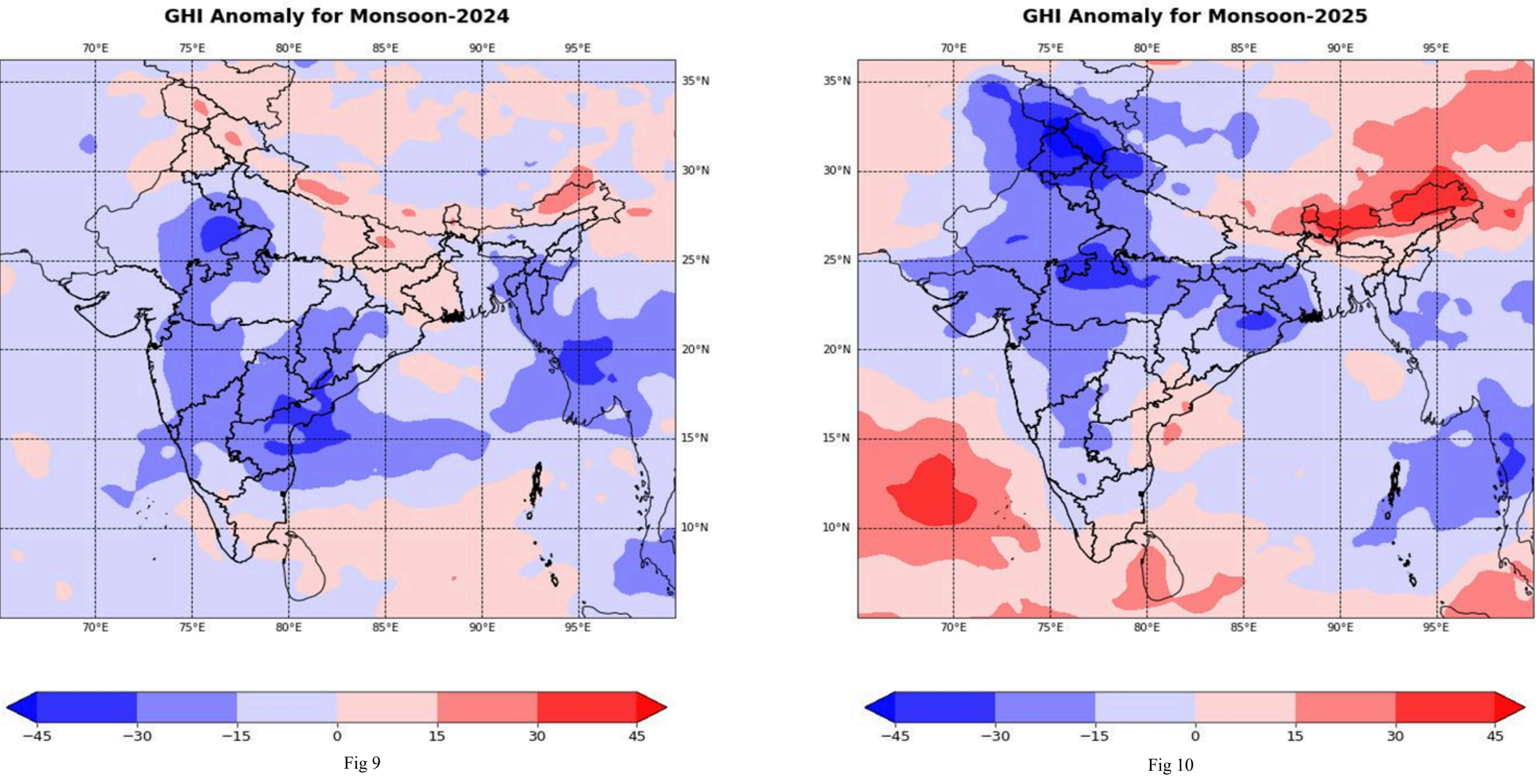
The standard deviation for 2024 and 2025 shows a pronounced negative anomaly across Gujarat and Rajasthan in 2025 compared to 2024, indicating that 2025 experienced a significantly wetter monsoon season than the previous year.



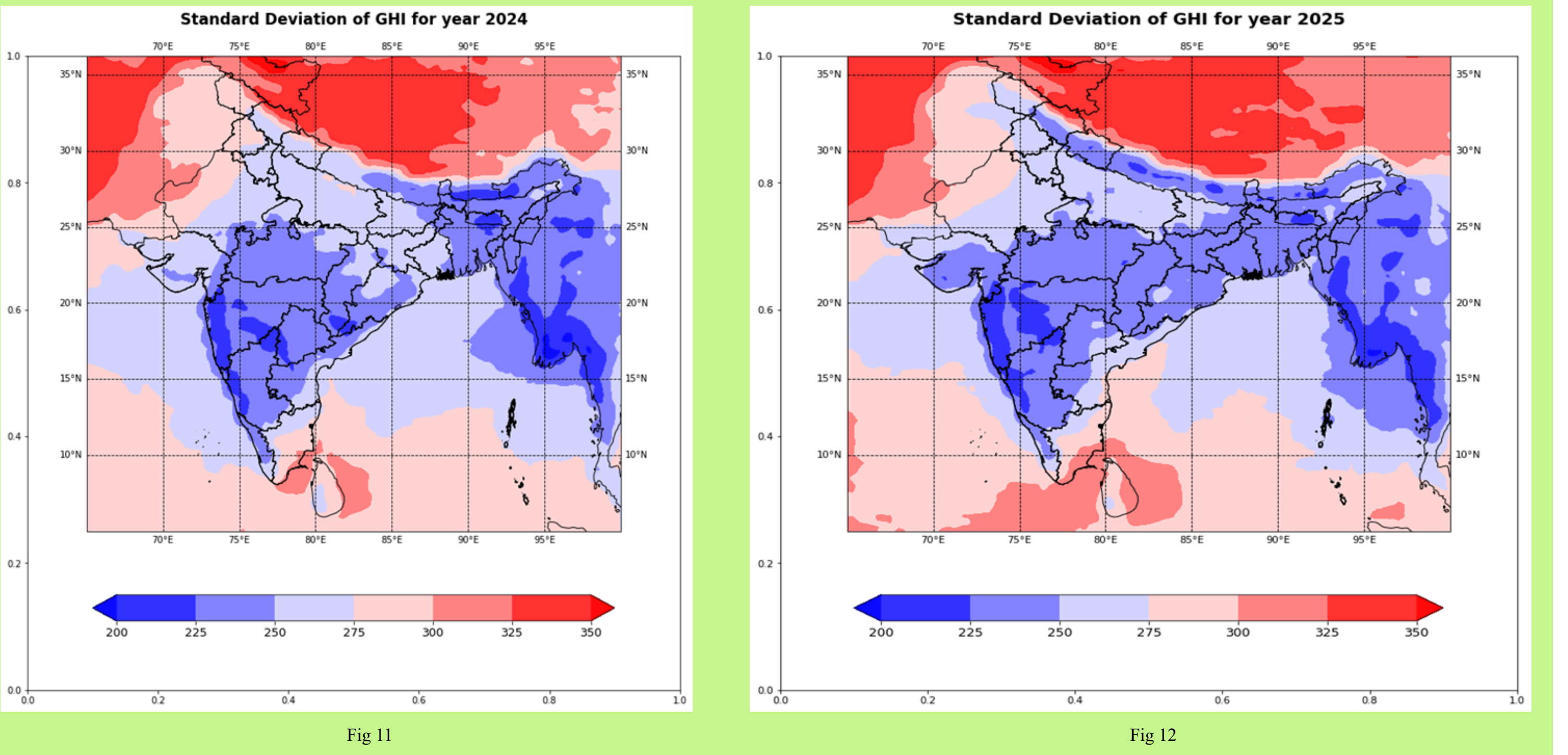
The yearly trend analysis for 30 years for the four locations in Khavda, Jaisalmer, Pavagada and Kamuthi are presented.

It has been observed that for all the locations, the long term trend for the GHI is declining.

This is due to the increase in cloud cover over the Indian region due to more weather disturbances and also due to the increasing water holding capacity of air associated with global warming.



The long-term GHI anomaly analysis for 2024 and 2025 shows a pronounced negative anomaly across Gujarat and Rajasthan in 2025 compared to 2024, indicating that 2025 experienced a significantly wetter monsoon season than the previous year.



Results and Discussion

The detailed analysis of GHI, wind speed at 100 m and rainfall for 30 years was done and the trend analysis was carried out for different locations as well as spatial trend over Indian region. It was observed that due to the formation of more extreme weather events and also due to climate change more cloud formation is happening so that the radiation trend in long term is decreasing. Wind speed also decreasing in certain regions.

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